North Edge CRM

Competitive Audit & Technology Benchmark Report

Version 3.0 • March 3, 2026

North Edge System • Toronto, Canada • www.northedgesystem.com

Classification: Internal – Engineering & Product

- Deep code-level CRM capability audit (51K+ backend LOC, 48K+ frontend LOC analysed)
- Competitive benchmarking against Top 8 CRM platforms
- Technology stack comparison and maturity scoring
- Functional feature roadmap: NOW / NEXT / LATER
- 27 unique differentiators and 8 gap areas identified (7 gaps closed since v2)
- AI capabilities deep-dive: RAG, 3-tier scoring, action orchestration, campaign AI
- Real-time infrastructure audit: SignalR hubs, event broadcasting, presence tracking
- **NEW**: Email integration (bidirectional M365/Gmail), Direct Chat, Quotes, Price Books

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Document Revision History

Version	Date	Author	Changes
1.0	February 11, 2026	Engineering	Initial competitive audit — 35 pages, 160 endpoints, 38 entities, 15 differentiators, 17 gaps
2.0	February 28, 2026	Engineering	Deep code re-audit — corrected to 47 pages, 199 endpoints, 83 entities, 20 differentiators, 15 gaps. Added Marketing & Campaigns (Full), SignalR implementation status, AI deep-dive, Stack Maturity Scorecard
3.0	March 3, 2026	Engineering	Major capability expansion — 53 pages, 254 endpoints, 88 entities, 27 differentiators, 8 gaps. **7 gaps closed**: Email Integration (bidirectional

			M365/Gmail), Quotes, Price Books, Attachments, Direct Chat, Import Progress. +27K LOC growth. Maturity score 59→72.

Current version: 3.0 — This document supersedes all previous versions.

1. Executive Summary

North Edge CRM is an Angular 21 + .NET Clean Architecture platform with **53 page components**, **43 backend API controllers (254 endpoints)**, **88 domain entities**, **7+ AI/ML service implementations**, and a full campaign management system with AI-powered recommendations. **Since Version 2**, the platform has added **bidirectional email integration** (Microsoft 365 + Gmail OAuth), **direct team chat** with SignalR, **quote/proposal generation**, **price book management**, **file attachments**, and **import with real-time progress tracking**.

The system features a proprietary CQVS lead scoring framework, a 3-tier AI fallback chain (Azure OpenAI → OpenAI → Rule-based), an AI assistant with RAG knowledge retrieval (Azure AI Search + Foundry Agent), a full Decision Engine with multi-step approval chains and SLA auto-escalation, campaign attribution with explainability, and a drag-and-drop dashboard with 18 widget types. **NEW**: Email inbox with full IMAP operations, team messaging, and opportunity quotes.

The backend totals **51,753 lines of C#** (+20% from v2) and the frontend comprises **~48,624 lines of TypeScript** (+18%) and **~59,325 lines of SCSS** (+24%).

The platform has advanced from **Upper Tier 2** to **approaching Tier 1** — unique AI capabilities now rival enterprise CRMs, and 7 major gaps have been closed. Key remaining gaps are: Visual Workflow Builder, Calendar Sync, Contract Management, and Territory Management. Direct competitors are **SugarCRM, Freshsales, Zoho CRM, and HubSpot** (mid-market).

2. CRM Capability Inventory

Dimension	Count	Detail
Frontend Page Components	**53**	+6 from v2 (2,532 LOC dashboard, 2,610 LOC lead form, mailbox pages, chat pages)
Routes (lazy-loaded)	**133**	+36 from v2 (67 CRM + 38 Supply Chain + 28 new)
Backend API Controllers	**43**	+5 from v2 (37 CRM + 6 SCM)
API Endpoints	**254**	+55 from v2 (GET 95, POST 78, PUT 35, DELETE 22, PATCH 18, misc 6)
Domain Entities	**88**	+5 from v2 (61 CRM + 27 SCM)
Application Service Interfaces	**37**	+11 from v2 (Clean Architecture layer)
Infrastructure Services	**45+**	Including 3 lead scoring + email sync + direct chat + attachments
Background Workers	**5**	+1 from v2 (Email, Notifications, Renewals, SLA Escalation, Import Progress)
AI/ML Implementations	**7+**	Chat/RAG, 3× lead scoring, Foundry, Search, Campaign AI, Email AI
MediatR Handlers	**8**	+2 from v2 (2 query + 6 event handlers)

SignalR Hubs	**2**	PresenceHub + CrmEventsHub (enhanced with chat events)
Permission Keys (RBAC)	**23**	+2 from v2 (+ 3 visibility scopes: Self/Team/All)
Dashboard Widget Types	**18**	16 cards + 2 charts
Settings Pages	**20+**	Roles, perms, tenants, policies, automation, email connections
EF Core Configurations	**52**	+5 from v2 (Entity type configs + 179 migrations)
Contract DTOs	**156**	+7 from v2 (In 27 subdirectories)
Backend LOC	**51,753**	+8,738 from v2 (+20% growth)
Frontend TS LOC	**~48,624**	+7,406 from v2 (+18% growth)
Frontend SCSS LOC	**~59,325**	+11,478 from v2 (112 SCSS files, +24% growth)

3. Feature Maturity Summary

Feature Area	Pages	Maturity	Key Highlights
Dashboard	1 (2,532 LOC)	Full	18 widgets, DnD, AI orchestration, forecast, coaching, manager health
Customers	2	Full	CRUD, bulk actions, CSV import, lifecycle/owner filters, metric cards
Contacts	2	Full	Full CRUD, CSV import, bulk assign/lifecycle, tabbed detail
Leads	3 (2,610 LOC form)	Full	CQVS scoring, 3-tier AI, cadence, conversion, duplicate check, import
Opportunities	7	Full	Stage mgmt, approvals, onboarding, review checklist, team, expansion, **quotes**
Activities	2	Full	Table/calendar/tasks views, type/status filters, overdue tracking
Decision Engine	5	Full	Inbox, multi-step chains, AI draft, SLA escalation, delegation, audit
Marketing & Campaigns	4 + 13 API	Full	CRUD, attribution with explainability, AI recommendations, health scoring
Email Integration	**8 routes**	**Full (NEW)**	**Bidirectional sync M365/Gmail, inbox/sent/drafts/trash/spam, templates, OAuth**
Direct Chat	**4 routes**	**Full (NEW)**	**Thread-based team messaging, SignalR real-time, typing indicators**
Quotes & Proposals	**3 routes**	**Full (NEW)**	**Quote generation, versioning, line items, PDF export**
Price Books	**3 routes**	**Full (NEW)**	**Product catalogs, tiered pricing, date validity**

Attachments	**3 routes**	**Full (NEW)**	**Multi-entity file upload/download, policy-based access**
Settings	20+	Full	Users, roles, perms, tenants, automation, qualification, dashboard packs, **email connections**
AI Assistant	Integrated	Full	Chat + RAG (AI Search + Foundry), action exec, undo (60s), risk tiers
Auth & Security	4 public pages	Full	JWT, RBAC (23 perms), invite flow, password reset, visibility scopes
Multi-Tenancy	Core infra	Full	Per-tenant policies, provisioning, feature gating, dashboard defaults
Real-Time	2 SignalR hubs	**Enhanced**	Presence + CRM events + **chat + import progress**, worker broadcasts
Automation	5 workers	Full	Renewal, SLA escalation, email queue, notification alerts, **import progress**

4. Unique Differentiators (27)

These are capabilities that most competitors either lack entirely or implement at a basic level:

#	Differentiator	Description
1	CQVS Lead Scoring Framework	Proprietary 6-factor model (Budget, Readiness, Timeline, Problem Severity, Economic Buyer, ICP Fit) with 21 contextual criteria
2	3-Tier AI Lead Scoring	Azure OpenAI → OpenAI → Rule-based automatic fallback. No competitor has multi-provider resilience
3	AI Execution Orchestration	Priority-scored action table with risk/urgency tiers and 60-second undo window on dashboard
4	AI Chat + RAG + Action Execution	Foundry Agent with Azure AI Search grounding, executes CRM actions with risk gating, review workflow, and undo
5	Cost-of-Not-Knowing Metric	Risk-weighted intelligence gap trend — unique KPI measuring data quality cost per deal
6	Truth Metrics Widget	Dashboard widget surfaces data integrity and confidence scoring across the pipeline
7	Confidence-Weighted Pipeline	Trust-scored deal values — no competitor weights pipeline by data quality confidence
8	Decision Engine + SLA Escalation	Multi-step approval chains with background worker auto-escalation, delegation, AI assist drafts
9	Expansion Signal Detection	Automated upsell/cross-sell identification from opportunity patterns with dedicated API
10	Renewal Automation Worker	Background service auto-creates renewal opportunities before contract expiry, runs every 12h

11	Qualification Policy Engine	Per-tenant rules with contextual threshold modifiers per deal type × segment × stage
12	DnD Dashboard (18 widgets)	More widget types than most Tier 2 dashboards with CDK drag-and-drop reordering + chart toggles
13	Manager Pipeline Health View	Coaching-oriented pipeline view with team performance overlay and truth gap detection
14	Priority Stream	Unified cross-entity feed combining leads, opportunities, activities in priority order with filtering
15	Lead Cadence Tracking	Evidence-source tracking with buyer engagement cadence scoring and SLA breach alerts
16	Campaign Attribution + Explainability	First-touch attribution model with evidence trail, candidate listing, and full explainability UI
17	Campaign AI Recommendations	Rule-based recommendations with confidence, impact estimates, evidence, accept/dismiss/snooze workflow
18	Campaign Health Scoring	Composite 0-100 health score with trend tracking, reason chips, and historical snapshots
19	SignalR Real-Time Broadcasting	All 5 workers publish events via SignalR to tenant + user channels; live presence tracking
20	Tenant Feature Gating	Per-tenant module toggle (e.g., supply chain, marketing) with route-level guard and disabled page
21	**Bidirectional Email Sync (NEW)**	**Full M365 + Gmail OAuth integration with IMAP operations, inbox/sent/drafts/trash/spam/starred/archive**
22	**Direct Team Chat (NEW)**	**Built-in SignalR-powered team messaging — eliminates need for Slack/Teams integration**
23	**Quote/Proposal System (NEW)**	**Quote generation with versioning, line items, approval workflow, PDF export**
24	**Price Book Management (NEW)**	**Product catalogs with tiered pricing, date validity, opportunity integration**
25	**Multi-Entity Attachments (NEW)**	**File attachments across all entities with policy-based access control**
26	**Import with Real-Time Progress (NEW)**	**CSV import with SignalR-powered progress bars and error reporting**
27	**Email templates (NEW)**	**Saved email templates with variables for quick personalized outreach**

5. Competitive Positioning vs. Top Leaders

Capability	North Edge	Salesforce	HubSpot	Dynamics	Zoho	Freshsales	Sugar
Contact/Account Mgmt	Full	Full	Full	Full	Full	Full	Full

Lead Mgmt & Scoring	Full + AI	Full + AI	Full + AI	Full + AI	Full	Full + AI	Full
Opportunity/Pipeline	Full	Full	Full	Full	Full	Full	Full
AI Lead Scoring	3-tier fallback	Einstein	Predictive	Copilot	Zia	Freddy	SugarP
AI Chat Assistant	Full + Actions	Einstein GPT	ChatSpot	Copilot	Zia	Freddy	—
AI Action Exec + Undo	Yes (60s)	Partial	No	Partial	No	No	No
Decision/Approval Engine	Full + SLA	Full	Basic	Full	Partial	No	Full
Dashboard Customization	18 widgets+DnD	Full	Full	Full	Full	Basic	Basic
RBAC (Granular)	23 perms	Full	Tiered	Full	Full	Basic	Full
Multi-Tenancy	Full	Full	No	Full	Full	No	Partial
CSV Import/Export	Full + Progress	Full	Full	Full	Full	Full	Full
Audit Trail	Full	Full	Partial	Full	Full	Basic	Full
Email Integration	**Full (bi)**	Full (bi)	Full	Full	Full	Full	Full
Campaign Management	Full	Full	Full	Full	Full	Full	Full
Marketing Automation	Partial	Pardot	Full	Full	Full	Full	Partial
Quote/CPQ	**Full (NEW)**	Full	Full	Full	Full	No	Full
Price Book	**Full (NEW)**	Full	Full	Full	Full	Basic	Full
Team Chat	**Full (NEW)**	Chatter	No	Teams	Cliq	No	No
Report Builder	Basic	Full	Full	Full	Full	Basic	Full
Web-to-Lead Forms	Missing	Full	Full	Full	Full	Full	Full
Mobile Native	Responsive	App	App	App	App	App	App

6. Technology Stack Benchmarks

Dimension	North Edge	Salesforce	HubSpot	Dynamics 365	Zo
Frontend	Angular 21	Aura/LWC	React	React+Fluent	Proprie
UI Library	PrimeNG 21	Lightning DS	Custom	Fluent UI	Custom

State Mgmt	Signals	LDS	Redux	Redux/Zustand	Custom
Backend	C# (.NET 10)	Java/Apex	Java+Node	C# (.NET)	Java
Architecture	Clean (4-layer)	MVC+Platform	Microservices	CQRS+Micro	Mono→
Database	SQL Server/EF	Oracle+Propr.	MySQL+HBase	SQL+Dataverse	Postgre
Auth	JWT+RBAC	OAuth+SAML+SSO	OAuth+SSO	AzureAD+RBAC	OAuth+
AI Runtime	AzureOAI+OAI+Rules	Einstein GPT	ChatSpot/OAI	AzureOAI/Copilot	Zia
AI Resilience	3-tier fallback	Single	Single	Single+fallback	Single
RAG/Knowledge	AI Search+Foundry	Data Cloud	Knowledge Base	Copilot Studio	—
Background Jobs	5 Hosted Svc	Batch Apex	Sidekiq+Kafka	AzFunc+SvcBus	Custom
Message Bus	Azure Svc Bus	Platform Events	Kafka	Azure Svc Bus	Custom
Email Service	**Graph+ACS+OAuth**	SF Email	SendGrid	Exchange+Graph	Custom
Hosting	Azure SWA+AppSvc	SF Cloud	AWS	Azure Cloud	Proprie
Multi-Tenancy	App-level	Platform-native	Account	Dataverse-native	Accoun
API Style	REST (254)	REST+SOAP+GQL	REST+GraphQL	REST+OData+GQL	REST+G
Real-time	SignalR (Enhanced)	Streaming API	WebSockets	SignalR	Long-p
Mobile	Responsive	iOS+Android	iOS+Android	iOS+Android+PWA	iOS+An
Extensibility	Code-level only	Apex+AppExch	WF+Marketplace	PowerPlat+Plugins	Deluge-

7. Stack Maturity Scorecard

Category	North Edge	Salesforce	HubSpot	Dynamics	Zoho	Freshsales	SugarCRM
Frontend Modernity	9/10	7/10	9/10	8/10	6/10	8/10	4/10
Backend Architecture	9/10	8/10	9/10	9/10	6/10	7/10	5/10
AI Sophistication	9/10	9/10	7/10	9/10	6/10	7/10	5/10
AI Resilience (Fallback)	10/10	6/10	4/10	7/10	4/10	4/10	4/10
Database/ORM	8/10	8/10	8/10	9/10	7/10	8/10	6/10
Auth & Security	7/10	10/10	8/10	10/10	8/10	7/10	7/10
API Design	7/10	10/10	9/10	10/10	8/10	7/10	7/10

Background Processing	8/10	9/10	9/10	9/10	6/10	7/10	4/10
Extensibility	4/10	10/10	8/10	10/10	7/10	6/10	6/10
Mobile	5/10	9/10	9/10	9/10	9/10	9/10	7/10
Real-time Capabilities	**8/10** (+3)	9/10	8/10	9/10	5/10	7/10	3/10
Email Integration	**8/10** (+3)	9/10	9/10	9/10	8/10	8/10	7/10
Average Score	**7.7** (+0.4)	8.6	8.0	8.6	6.5	7.0	5.3

Stack Strengths (Updated)

- **Angular 21 + PrimeNG 21** — Latest framework version. SugarCRM is still on Backbone (legacy).
- **Clean Architecture (4-layer)** — Architecturally aligned with Dynamics 365.
- **3-Tier AI Fallback (10/10)** — No other CRM has multi-provider AI resilience.
- **Bidirectional Email (NEW)** — Full M365 + Gmail OAuth with IMAP operations.
- **Real-time Chat (NEW)** — Built-in team messaging via SignalR.
- **Azure Service Bus** — Same enterprise messaging as Dynamics 365.

Stack Gaps (Reduced)

- ~~Email Integration~~ **CLOSED** — Now full bidirectional with M365/Gmail OAuth
- ~~Quote/CPQ~~ **CLOSED** — Now has quote generation with PDF export
- No SSO/SAML/OAuth2 provider — Still blocks enterprise procurement
- No GraphQL or OData — REST-only limits query flexibility
- No native mobile app — Responsive web + Flutter in development
- No plugin/extension system — Can't build marketplace ecosystem

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8. Feature Gaps (8)

7 gaps closed since Version 2. Remaining gaps:

Severity	Feature Gap	Details
HIGH	Visual Workflow Builder	Decision Engine exists, but no drag-and-drop automation designer. Table stakes for enterprise.
MEDIUM	Calendar Sync (Google/Outlook)	Internal calendar view exists; no native Google/Outlook sync yet.
MEDIUM	Contract Management	Renewal worker + quotes exist but no formal contract entity/lifecycle workflow.
MEDIUM	Territory Hierarchy	No territory tree structure for regional access control.
MEDIUM	Revenue Forecasting	Confidence-weighted pipeline exists but no formal forecast periods or targets.
LOW	Customer Self-Service Portal	No customer-facing portal for support/orders.

LOW	Full CPQ Configurator	Price lists + quotes exist; product configurator missing.
LOW	Scheduled Reports & Digests	No daily/weekly email digests or report subscriptions.

Gaps Closed Since v2

Gap	Status	Evidence
Email Integration (Bidirectional)	✅ CLOSED	MailboxSyncService (1,155 LOC), MailboxController (430 LOC), M365+Gmail OAuth
Product & Price Book	✅ CLOSED	PriceListsController (148 LOC) with full CRUD
Quote/Proposal Generation	✅ CLOSED	OpportunityQuotesController (252 LOC) with versioning and PDF
File Attachments	✅ CLOSED	AttachmentsController (254 LOC) with policy-based access
Import Progress Tracking	✅ CLOSED	ImportJobsController + ImportJobRealtimeProgressWorker with SignalR
Notes & Comments	✅ CLOSED	Direct chat threads provide team collaboration
Real-time Team Communication	✅ CLOSED	DirectChatService (389 LOC) with SignalR messaging

9. Functional Roadmap: NOW / NEXT / LATER

NOW (0–3 months) — Deal-Breakers That Block Sales

#	Feature	Why NOW
N1	Visual Workflow Builder	Enterprise buyers expect drag-and-drop automation. Competes with Salesforce Flow.
N2	Calendar Sync (Google/Outlook)	Completes the communication-native vision. Reps need meeting visibility.
N3	SSO/SAML Enterprise Auth	Enterprise deal-blocker. IT teams require SSO before procurement approval.
N4	Webhook/Event System	Can't integrate with Zapier, Make, N8N without webhooks. Critical for ecosystem.

NEXT (3–6 months) — Competitive Parity

#	Feature	Why NEXT
X1	Contract Management	Quotes now exist; need contract entity/lifecycle to close the loop.
X2	Territory Management	Enterprise orgs with 50+ reps need territory-based access.

X3	Revenue Forecasting Engine	Dashboard metrics exist; add target-vs-actual tracking, period management.
X4	Scheduled Reports & Digests	Managers expect morning pipeline emails and weekly summaries.
X5	Mobile App (Flutter)	Flutter app in separate repo; needs MVP launch for field sales.

LATER (6–12 months) — Differentiation & Tier 1 Push

#	Feature	Strategic Value
L1	Full Marketing Automation	Add nurture sequences, engagement scoring to existing campaigns.
L2	Customer Portal	Self-service for customers — extends multi-tenancy.
L3	Case/Support Ticketing	Expands from Sales CRM to full CRM (sales + service).
L4	Plugin/Extension Framework	Marketplace ecosystem for partners and customers.
L5	Advanced Analytics & BI	Cohort analysis, pivot tables, drill-down — Power BI rival.
L6	AI Conversation Intelligence	Call transcription, sentiment analysis, coaching insights.
L7	Native Mobile App (iOS+Android)	Full native experience with offline support.

10. Dependency Chain (Build Order)

```
→ Visual Workflow Builder (NOW) → Plugin Framework (LATER)
→ Calendar Sync (NOW) → Full Marketing Automation (LATER)
→ SSO/SAML (NOW) → Enterprise Customers
→ Webhooks (NOW) → Third-party Integrations → Plugin Framework (LATER)
→ Contract Mgmt (NEXT) → Customer Portal (LATER)
→ Territory Mgmt (NEXT) → Forecasting Engine (NEXT)
→ Mobile App MVP (NEXT) → Native Mobile (LATER)
→ Case Ticketing (LATER) → Customer Portal (LATER)
```

Critical Path: Visual Workflow Builder → SSO/SAML → Webhooks → Mobile App MVP

These 4 features unblock the most downstream work and close the biggest competitive gaps. Email, Quotes, and Price Books are now complete — the core deal-closure workflow is fully inside the CRM.

11. Immediate Competitors

Tier	CRMs	Rationale
Tier 1 (Enterprise)	Salesforce, Dynamics 365, Oracle CX	Full platform: marketing automation, CPQ, service, analytics, marketplace
Tier 2 Upper (YOU)	**North Edge CRM**, SugarCRM, Freshsales	Strong core CRM + AI + campaigns + **email + quotes + chat**. Missing workflow builder.
Tier 2 Lower	Zoho, Pipedrive, Monday CRM	Broader ecosystem but weaker on AI/decision engine

Tier 3 (Basic)	Insightly, Capsule, Less Annoying CRM	Basic contact/deal management only

Competitive Position Change: North Edge now has email integration, quotes, price books, and team chat — **exceeding SugarCRM, Freshsales, and Zoho** on multiple fronts. AI and decision-engine capabilities exceed all three. Remaining gaps vs. Tier 1: workflow builder, mobile app, third-party marketplace.

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12. Real-Time Capabilities Analysis

North Edge CRM now scores **8/10 on real-time capabilities** — up from 5/10 in v2. Two SignalR hubs (PresenceHub + CrmEventsHub) are fully implemented, **direct team chat with real-time messaging**, **import progress streaming**, and all 5 background workers publish events via SignalR.

Capability	Current Status	Detail
SignalR Hubs	Implemented (2)	PresenceHub (join/leave broadcast) + CrmEventsHub (tenant/user groups)
Event Publisher	Implemented	SignalRCrmRealtimePublisher — ICrmRealtimePublisher interface, tenant+user scoped
Presence Tracking	Implemented	PresenceHub with user join/leave, online status broadcasting
Worker → Browser Push	Implemented (5)	All 5 workers call PublishTenantEventAsync/PublishUserEventAsync
Push notifications	Implemented	NotificationAlertWorker pushes SLA breaches, idle deals, coaching alerts
Direct Team Chat	**Implemented (NEW)**	**Real-time messaging via SignalR with typing indicators**
Import Progress	**Implemented (NEW)**	**CSV import progress streamed to browser via SignalR**
Live record updates	Not yet	No record-level change broadcasting to other viewers
Streaming AI chat	Not yet	Full response wait — not streaming tokens via IAsyncEnumerable
Live dashboard	Partial	Some widgets auto-refresh; full SignalR deltas pending

CRM	Score	Technology
Salesforce	9/10	Streaming API, Platform Events, Pub/Sub API, real-time record locking
Dynamics 365	9/10	SignalR-based real-time updates, Power Automate triggers, live co-authoring
HubSpot	8/10	WebSocket-based live updates, real-time activity feed, live chat
North Edge CRM	**8/10** (+3)	**SignalR (2 hubs, chat, import progress, presence, worker broadcasts)**
Pipedrive	7/10	WebSocket live updates on deals/activities, real-time sync
Freshsales	7/10	WebSocket updates, real-time lead tracking, live chat

Zoho	5/10	Partial real-time via long-polling, Zoho Cliq integration
SugarCRM	3/10	Minimal — mostly polling-based

13. SignalR Implementation Status

The SignalR infrastructure is fully operational and enhanced since v2. Two hubs handle presence and CRM events. A dedicated publisher (SignalRCrmRealtimePublisher) provides tenant-wide and user-specific broadcasting. All 5 background workers actively push events. **NEW**: Direct chat and import progress use SignalR streams.

Status	Component	Detail
BUILT	PresenceHub	User join/leave broadcasting, online status tracking via groups
BUILT	CrmEventsHub	Tenant-scoped and user-scoped event groups, general broadcasting
BUILT	SignalRCrmRealtimePublisher	ICrmRealtimePublisher impl — PublishTenantEventAsync, PublishUserEventAsync
BUILT	EmailQueueWorker → SignalR	Pushes email delivery status (sent/failed) to tenant + user channels
BUILT	NotificationAlertWorker → SignalR	Pushes SLA breach, idle deal, coaching alerts to user channels
BUILT	DecisionSlaEscalationWorker → SignalR	Pushes escalation events to approver's user channel
BUILT	RenewalAutomationWorker → SignalR	Pushes renewal creation events to opportunity owner's channel
BUILT	**ImportJobRealtimeProgressWorker (NEW)**	**Pushes CSV import progress to user channel in real-time**
BUILT	**DirectChatService → SignalR (NEW)**	**Real-time message delivery, typing indicators, read receipts**
REMAINING	AI Assistant Streaming	Switch to IAsyncEnumerable + SignalR stream for token-by-token UX
REMAINING	Dashboard Live Metrics	Entity change detection → broadcast widget deltas to viewers
REMAINING	Pipeline Kanban Live	Broadcast opportunity stage changes for live card moves
REMAINING	Record-Level Presence	Show who is viewing/editing a specific record in real-time

14. AI Capabilities Deep-Dive

North Edge CRM has **7+ distinct AI implementations** — more than most Tier 2 competitors and rivaling Tier 1 platforms in several areas. The AI architecture features multi-provider resilience, RAG grounding, action execution with risk-based review, and campaign intelligence.

#	Implementation	LOC	Key Capabilities
1	AssistantChatService	1,080	Foundry Agent with RAG (AI Search), action execution with risk tiering (low=auto, medium/high=review), 60s undo
2	AzureOpenAILeadScoringService	126	Primary AI scorer — structured JSON response with score (0-100), confidence (0-1), reasoning
3	OpenAILeadScoringService	117	Secondary fallback — identical prompt/response schema. Auto-activates when Azure OpenAI unavailable
4	RuleBasedLeadScoringService	33	Tertiary fallback — additive scoring (base 20, max 100), fixed confidence 0.35. Ensures scoring never fails
5	FoundryAgentClient	224	Azure AI Foundry Agents HTTP client with thread-based conversation, polling with rate-limit retry
6	AzureSearchKnowledgeClient	87	RAG retrieval layer — configurable top-k, filter expressions, content truncation
7	MarketingService (AI)	1,232	5 AI recommendation types with confidence, impact estimates, evidence
8	**MailboxSyncService (AI) (NEW)**	**1,155**	**Email summarization, thread analysis, suggested responses**

AI Architecture Highlights

- **3-Tier Scoring Fallback (10/10 Resilience)** — Azure OpenAI → OpenAI → Rule-based. No other CRM has automatic multi-provider AI failover.
- **Action Risk Tiering** — Low-risk actions auto-execute. Medium/high-risk go to review. Confidence < 0.55 forces review regardless of risk level.
- **60-Second Undo Window** — Every AI-executed action can be undone within 60 seconds. Unique in the CRM market.
- **RAG Grounding** — Azure AI Search retrieves knowledge documents; Foundry Agent uses them for context-aware responses.
- **Campaign Intelligence** — 5 recommendation types with confidence scoring, impact estimates, and evidence.
- **Email AI (NEW)** — Email summarization and thread analysis for inbox prioritization.

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15. New Capabilities Since v2

Email Integration (MailboxSyncService + MailboxController)

Total LOC: 1,585

Feature	Description
OAuth Authentication	Microsoft 365 and Gmail OAuth flows for secure email access
Bidirectional Sync	Full IMAP operations — inbox, sent, drafts, trash, spam, starred, archive

Email Send	Compose and send emails directly from CRM with tracking
Email Templates	Saved templates with variable substitution for quick outreach
Email Threading	Thread-based conversation view with contact/lead/opportunity linking
AI Summarization	Automatic email summarization for inbox prioritization

Direct Team Chat (DirectChatService + DirectChatController)

Total LOC: 543

Feature	Description
Thread-based Messaging	Conversation threads between team members
SignalR Real-time	Instant message delivery without polling
Typing Indicators	See when colleagues are typing
Read Receipts	Know when messages have been read
User Presence	Online/offline status integration with PresenceHub

Quotes & Price Books

Total LOC: 400

Feature	Description
Price Lists	Product catalogs with tiered pricing and date validity
Quote Generation	Create quotes from opportunities with line items
Quote Versioning	Track quote revisions with approval workflow
PDF Export	Generate professional quote PDFs for customers

Attachments System

Total LOC: 254

Feature	Description
Multi-entity Support	Attach files to leads, opportunities, customers, contacts, activities
Policy-based Access	Respect RBAC permissions for file access
Upload/Download	Secure file upload and download with progress tracking

16. Recommendations

Immediate Priorities (This Quarter)

- **Ship Visual Workflow Builder** — Enterprise deal-blocker. Drag-and-drop automation is table stakes. Leverage existing Decision Engine infrastructure.
- **Add SSO/SAML** — IT teams require SSO before procurement approval. Azure AD B2C or IdentityServer integration.
- **Export Webhooks** — Leverage existing Azure Service Bus. Publish entity events → outbound webhook delivery. Enables Zapier/Make/N8N integrations.
- **Calendar Sync** — Google Calendar + Outlook Calendar integration to complete communication-native vision.

Technical Quick Wins

- **Stream AI Chat via SignalR** — Switch AssistantChatService to IAsyncEnumerable + SignalR stream. Token-by-token UX with no extra infrastructure.
- **Add Record-Level Presence** — Infrastructure exists (2 hubs, publisher). Add record viewing/editing presence.
- **Add OData or GraphQL layer** — Start with OData on key entities (read-only) for power users and integrators.

Strategic Positioning

North Edge CRM has **closed 7 major gaps since v2** and now has distinctive advantages that no Tier 2 competitor can match:

- **Email integration** with M365/Gmail OAuth (SugarCRM, Freshsales have no native OAuth)
- **Direct team chat** built-in (eliminates Slack/Teams requirement)
- **AI-first architecture** with 3-tier fallback (no competitor has this resilience)
- **Full quote/proposal system** with price books

The strategy should be: Close the remaining enterprise blockers (Workflow Builder, SSO, Webhooks), then lean into AI differentiators as the competitive moat.

Position as: "The AI-first CRM with enterprise-ready communication" — now that email, chat, quotes, and price books are shipped, this claim is fully defensible.

Roadmap Summary

Phase	Features	Timeline	Outcome
NOW	4 features	0–3 months	Close enterprise blockers. Visual workflow, SSO, webhooks, calendar
NEXT	5 features	3–6 months	Full Tier 2 parity. Contract mgmt, territories, forecasting, mobile MVP
LATER	7 features	6–12 months	Push into Tier 1. Marketing automation, portal, ticketing, analytics
TOTAL	**16 features**	**12 months**	**From Upper Tier 2 → Lower Tier 1**

End of Report

North Edge CRM Competitive Audit & Technology Benchmark • Version 3.0

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